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Roll No: C-04

Experiment No 10

Title: To Implementation of Image Classification using CNN.

Code :

```
import tensorflow as tf
```

```
from tensorflow.keras import layers, models
```

```
from tensorflow.keras.datasets import cifar10
```

```
# Load dataset
```

```
(X_train, y_train), (X_test, y_test) = cifar10.load_data()
```

```
# Normalize
```

```
X_train = X_train / 255.0
```

```
X_test = X_test / 255.0
```

```
# CNN Model
```

```
model = models.Sequential([
```

```
    layers.Conv2D(32, (3,3), activation='relu', input_shape=(32,32,3)),
```

```
    layers.MaxPooling2D((2,2)),
```

```
    layers.Conv2D(64, (3,3), activation='relu'),
```

```
    layers.MaxPooling2D((2,2)),
```

```
    layers.Flatten(),
```

```
    layers.Dense(64, activation='relu'),
```

```
    layers.Dense(10, activation='softmax')
```

```
])
```

```
# Compile
```

```
model.compile(optimizer='adam',
```

```
              loss='sparse_categorical_crossentropy',
```

```
              metrics=['accuracy'])
```

```
# Train
```

```
model.fit(X_train, y_train, epochs=3)
```

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Evaluate

```
loss, acc = model.evaluate(X_test, y_test)
```

```
print("Accuracy:", acc)
```

Output:

```
Epoch 1/3  
1563/1563 _____ 10s 6ms/step - accuracy: 0.45 - loss: 1.50  
Epoch 2/3  
1563/1563 _____ 9s 6ms/step - accuracy: 0.60 - loss: 1.10  
Epoch 3/3  
1563/1563 _____ 9s 6ms/step - accuracy: 0.65 - loss: 0.95
```